👀 What is Frontend?

Everything you see:

The visual part of a website or app that users interact with directly. This includes the **buttons** you click, the **colors** that guide your eye, and the overall **layout**.

"What are the main reasons you like a specific app or website?"

🛠️ Building Blocks

🧱

HTML

Structure

🎨

CSS

Styling

⚡

JavaScript

Actions

📄 HTML (HyperText Markup Language)

What is it?

- It’s **not** a programming language.
- Creates the fundamental **structure and skeleton** of the webpage.

How is it used in websites?

📝 **Headings**

📄 **Paragraphs**

🖼️ **Images**

🔗 **Links**

🏗️ Basic HTML Structure

```html
<!DOCTYPE html>
<html>
<head></head>
<body>
<!-- Visible content -->
</body>
</html>
```

<!DOCTYPE html>: Tells the browser the content is HTML.

<html>: The root element of the HTML document.

<head>: Contains meta-information (title, styles, scripts).

<body>: Contains the visible content (text, images, buttons).

The standard required for every HTML document.

🏷️ HTML Tags

Structure of a Tag

```
<tagname> Content </tagname>
```

`<h1> to <h6>`
Headings: Range from largest to smallest size.

`<p>`
Paragraph: Defines a block of text.

`<br> & <div>`
Layout: Line breaks and structural containers.

`<button>`
Action: Creates clickable buttons for user interaction.

`<select> & <option>`
Dropdowns: Creates menus and defines individual choices within them.

Tags tell the browser exactly what type of content to display.

💻 HTML Code Example

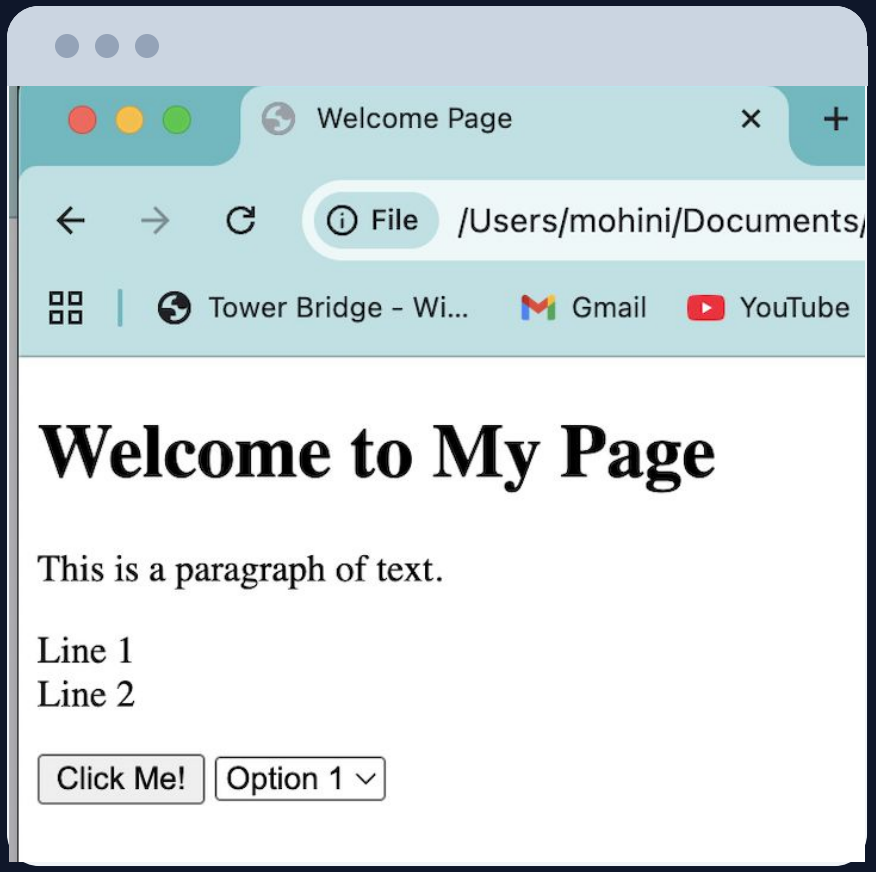
```html
<h1>Welcome to My Page</h1>

<p>This is a paragraph of text.</p>

<p>Line 1<br>Line 2</p>

<button>Click Me!</button>

<select>
<option>Option 1</option>
<option>Option 2</option>
</select>
```



🛠️ Exercise: HTML Elements

```html
<h2>🎧 Choose Your Mood Music</h2>

<select id="musicSelect">

<option>Happy 😊</option>
<option>Relax 😌</option>
<option>Workout 💪</option>
</select>

<button onclick="playMusic()">Play Music</button>
```

Practice creating a functional dropdown menu and an interactive button.
https://codepen.io/pen/

🎨 CSS (Cascading Style Sheet)

The visual engine that **transforms** HTML structure into design.

Empowers you to make webpages pretty and organized.

Key Styling Capabilities:

Typography
Fonts, colors, spacing, and sizing.

Backgrounds
Solid colors, gradients, and images.

Layout
Borders, padding, and positioning.

UI Elements
Buttons, inputs, and hover effects.

📋 CSS Syntax & Format

```
h1 { color: blue; font-size: 12px;
}
```

The Selector

The h1 part points to the specific HTML element you want to style on your page.

The Declaration Block

Everything inside the { } contains styles, separated by semicolons.

Proper syntax ensures your designs are applied correctly to the browser structure.

📝 Exercise: Style Analysis

```
/* Heading style */
h2 {
color: teal;
}

/* Dropdown style */
select {
font-size: 16px;
padding: 5px;
}

/* Button style */
button {
background-color: orange;
color: white;
padding: 5px 10px;
border: none;
}
```

Review the CSS selectors and their properties to identify how each element will be transformed.

⚡ JavaScript

The Logic Engine

JavaScript is a powerful Programming Language (PL) that breathes life into static pages.

- Makes websites **interactive and dynamic**.
- Responds to user actions like clicks and form inputs.

```
// Example Action
function musicSelection() {
var mood =
document.getElementById("musicSelect").value;
alert("You selected: " + mood);
}
```

🎮 The Web Trifecta: HTML + CSS + JS

HTML
Builds the structure 🏗️

CSS
Adds style and design 🎨

JavaScript
Adds actions and logic 🧠

Together, they create a **complete and interactive website** 🌐